PartsCompare

Universal Enterprise Sourcing Intelligence & BOM Governance Engine

Industry Partner: AMGC Airconditioning Services
Institution: ICCT Colleges • CEDT Nexus
Version: Enterprise Edition 2.4 • November 2026

🔐 Official Role Credentials & 2FA Anti-Impostor Reference

All accounts are protected by multi-factor authentication. Each staff member has a **unique, domain-relevant security question**, a **dedicated 6-digit recovery PIN**, and universal demo compatibility.

Staff Role	Assigned Personnel	Username	Password	6-Digit PIN	Unique Security Question & Answer
BOM Preparer FIELD OPERATIONS	Christian Dave D. Guiyab	`dave_pm`	`demo`	`240811`	**Q:** *What is your employee ID badge or favorite HVAC tool?* **A:** `manifold gauge`
Catalog Maintainer DATA OPERATIONS	Pearl Alexa J. Catequista	`pearl_dba`	`demo`	`350922`	**Q:** *What was the name of your first elementary school?* **A:** `san roque elementary`
System Administrator EXECUTIVE 2FA	Karen Reyes	`karen_admin`	`demo`	`890415`	**Q:** *What city or municipality were you born in?* **A:** `manila`

🛡️ Module 1: Dual-Access Gateway (Live Database vs. Safe Simulation)

🎯 Primary Purpose & Problem Solved

Prevents external evaluators, students, or unauthorized users from inspecting or corrupting private company quotation data, while providing an unrestricted, fully interactive demonstration.

📋 **Step-by-Step Operation:**

1. On the Landing Page (`#/`), choose between **"Sign In to Staff Workspace (Live Enterprise)"** or the 1-click **"Safe Simulation Sandbox"** roles (Karen, Pearl, or Dave).
2. In Simulation mode, all state writes strictly to isolated `sessionStorage`. Supabase Cloud synchronization is automatically bypassed.
3. A persistent 🎮 `SIMULATION DEMO SANDBOX ACTIVE` banner appears across all views.
4. Click **"Exit Sandbox"** at any time to return cleanly to the live home screen.

> 💡 ***Real-World Scenario:*** *A college professor evaluates the sourcing engine during project grading. The professor can create BOMs and edit prices freely in the Dave/Pearl simulation without altering AMGC's real contractor costs in the Supabase production database.*

🔑 Module 2: Anti-Impostor Password Recovery (Forgot Password)

🎯 Primary Purpose & Problem Solved

Enables instant self-service account recovery without external email server dependencies, while preventing brute-force takeovers through 5-attempt progressive lockouts and cryptographic audit logging.

📋 Step-by-Step Operation:

1. Navigate to `#/login` and click **"Forgot password?"**.
2. **Step 1:** Enter registered username (e.g. `pearl_dba`) and click *Continue →*.
3. **Step 2:** The system retrieves that user's unique security challenge. Enter the 6-Digit PIN and/or Security Answer, along with the desired new password (min 4 characters).
4. **Step 3:** Submitting with both factors (2FA) triggers immediate credential recovery (`IMMEDIATE_ADMIN_RESET`), logging the user in instantly and recording an audit trail.

> 💡 ***Real-World Scenario:*** *Dave is on a maintenance call in Cainta and forgets his account password. He opens the portal on his mobile tablet, verifies his identity with his PIN and secret answer "manifold gauge", and recovers access immediately without waiting for an administrator.*

🤖 Module 3: Offer Catalog Curation & AI Smart Quote Ingest (`#/catalog`)

3A. 🤖 AI Smart Quote Ingest (Natural Language Viber & Email Parser)

Purpose: Ingests unstructured, messy quotations received via Viber chats, WhatsApp messages, supplier quotation emails, or copied Excel ranges—eliminating manual data entry.

📋 **Step-by-Step Operation:**

1. Click the teal-blue gradient **"🤖 AI Smart Quote Ingest"** button in the Offer Catalog Workspace.
2. Paste raw text or click one of the 1-click demo chips (*Viber Chat*, *Supplier Email*, or *Excel Table*).
3. Click **"⚡ Run AI Analysis & Extract Quotes"**. The AI algorithm detects supplier headers, matches HVAC component descriptions via semantic synonyms, parses Philippine Peso prices (₱, php, pesos), and captures stock.
4. Review the confidence badges in the interactive table, edit any values inline, and click **"✓ Import & Synchronize"** to push to the live catalog.

> 💡 ***Real-World Scenario:*** *Everwell Parts texts Dave an informal Viber message: "CBB65 45uF 450V capacitor round - ₱405.00 each (45 pcs on hand)". Dave pastes the text into PartsCompare; the AI automatically normalizes it to standardized SKU `CBB65-45UF-450V`, extracts ₱405.00, and updates stock to 45 units in 3 seconds.*

3B. 📁 Upload Multi-Supplier CSV & CSV Template Generator

Purpose: Bulk-updates hundreds of supplier component prices simultaneously using structured spreadsheets, and provides a pre-formatted template for new vendor onboarding.

📋 **Step-by-Step Operation:**

1. Click **"CSV Template"** to download the standardized schema (`part_id`, `description`, `source`, `unit_price`, `stock`).
2. Click **"Upload Multi-Supplier CSV"** to select a populated price sheet.
3. The smart upsert engine auto-registers missing parts and vendors, updates prices, and timestamps verified dates within the 14-day freshness window.

> 💡 ***Real-World Scenario:*** *Manco Refrigeration issues their monthly wholesale price bulletin. Maintainer Pearl uploads `manco_nov_2026.csv`, refreshing quotes across 50 components in a single click.*

3C. ➕ Add Curated Component Offer & Dedicated Vendor Portal

Purpose: Allows maintainers to record single dated offers with invoice proof, or manage individual suppliers via drag-and-drop in the *Supplier CSV Update Dashboard*.

📋 **Step-by-Step Operation:**

1. Click **"+ Add New Offer"** to curate a single component quote from an accredited vendor.
2. Or switch to **"Supplier CSV Update Dashboard"** tab, pick an accredited vendor (Everwell, C&C, or Manco), and drag price sheets directly into their dedicated dropzone.

> 💡 ***Real-World Scenario:*** *C&C Airconditioning Supply calls with a special flash discount on defrost contactors (₱340.00). Dave logs the single offer with invoice code `#CC-FLASH-11` without needing a full spreadsheet.*

💡 Module 4: Interactive Feature Guide Hover Card System

🎯 Primary Purpose & Problem Solved

Provides zero-friction onboarding during defense presentations. When the cursor lands on any major action button or card, an animated guide card glides in showing its **Function**, **4-Step Workflow**, and **Real-World Scenario**.

📋 **Step-by-Step Operation:**

1. Hover the mouse cursor over **"🤖 AI Smart Quote Ingest"**, **"Upload Multi-Supplier CSV"**, or any workspace tab.
2. After a gentle 180ms dwell, an interactive guide card pops up with edge-detection (flips above if near the screen bottom).
3. Move the mouse into the card to read comfortably; the card remains open until the cursor exits.

📋 Module 5: BOM Preparer Workspace & Multi-Objective Sourcing (`#/lists`)

5A. Dynamic Sourcing Strategies & MCDA Optimization

Purpose: Evaluates bill of materials component lines against accredited supplier quotations using four algorithmic procurement strategies:

Sourcing Strategy	Algorithmic Principle	Optimal HVAC Scenario
1. Lowest Cost	Greedy unit price minimization per line item.	Standard routine repair where minimizing direct material budget is paramount.
2. Supplier Consolidation	Maximizes total line items fulfilled by a single accredited supplier to minimize logistics/freight costs.	Urgent turnaround overhaul where picking up all parts from one vendor saves critical technician travel time.
3. Resilience (Stock Buffer)	Selects candidates with the highest recorded warehouse stock headroom over project requirements.	High-volume commercial projects where inventory shortfalls cause expensive jobsite delays.
4. Balanced (Dynamic MCDA)	Multi-Criteria Decision Analysis weighting Cost (40%), Stock (30%), Freshness (20%), and Consolidation (10%).	Enterprise procurement balancing price competitiveness, vendor reliability, and data recency.

> 💡 ***Real-World Scenario:*** *For Job #AC-2026-088 (Split-Type AC Overhaul), Dave switches from Lowest Cost (split between 2 vendors) to Supplier Consolidation. C&C fulfills all required parts, reducing delivery fees and technician pickup delays.*

🔒 Module 6: 14-Day Freshness Rule & Preserved Sourcing Summaries (`#/summaries`)

6A. Governance Rule: Offer Freshness Policy (14 Calendar Days)

Supply chain prices in the Philippines fluctuate rapidly. PartsCompare strictly flags any quote verified more than 14 days ago as `EXCLUDED: STALE QUOTE`. Stale quotes cannot be signed off until re-verified.

6B. Governance Rule: "Unresolved is Never ₱0.00"

In incomplete BOMs, assuming an unquoted part costs zero leads to dangerous budget underestimation. PartsCompare excludes unresolved lines from the subtotal and flags the job sheet until an accredited quote is verified.

6C. Preserved Sourcing Summaries & Tamper-Proof Snapshots

Clicking **"Freeze Sourcing Summary"** cryptographically locks the agreed prices, chosen vendors, and rule settings. Future catalog edits will never alter historical sign-offs, guaranteeing 100% audit integrity.

> 💡 ***Real-World Scenario:*** *Six months after completing Job #AC-2026-088, the building owner audits the billing. Karen opens the frozen summary; the exact ₱3,795.00 breakdown signed off by Dave on 11/4/2026 remains preserved verbatim.*

👑 Module 7: Enterprise Administration Console (`#/admin`)

🎯 Primary Purpose & Governance Controls

Enables executive oversight by Administrator Karen: dynamic MCDA weight sliders, company profile branding (business name, currency PHP, contact), accredited supplier registry, and real-time audit ledger.

📋 **Step-by-Step Operation:**

1. Sign in as `karen_admin` → navigate to `#/admin`.
2. Adjust the **MCDA Utility Sliders** (Cost, Stock Buffer, Freshness, Consolidation) to adapt to market conditions.
3. Review the **Tamper-Evident Audit Event Ledger** recording every login, simulation, and quote ingest.
4. Approve or reject pending staff account registration requests.